Ludan Alsoudani

442-207-8985 | ludan.alsoudani@princeton.edu | linkedin.com/in/ludan-alsoudani/ | https://github.com/ludancom

Education

Princeton University Princeton, NJ

B.S.E in Computer Science, Minor in Robotics *Expected May 2028*

Relevant Coursework: Data Structures & Algorithms, Programming Systems, Intro to Machine Learning, Advanced Programming Techniques, Logic Design, Computer System Design, Reasoning About Computation

Technical Skills

Languages: Python, Java, Go, C/C++, JavaScript, HTML/CSS
Frameworks & Libraries: React, Node.js, Flask, FastAPI, pandas, NumPy, Matplotlib
Databases & Cloud: PostgreSQL, MySQL, AWS, Supabase, Neon, Render
Developer Tools & APIs: Git, GitHub, VS Code, OpenAI API, Anthropic API

Experience

StarQube Software Engineering Intern Jun. 2026 - Aug. 2026

StarQube *Geneva, Switzerland*

- Improved execution speed by 10% and significantly reducing stack memory overhead across production level financial computation pipelines by Implementing coroutine based execution models to replace legacy systems.
- Developing and optimizing components of StarQube's proprietary Financial Query Language (FQL) execution engine using modern C++20.
- Collaborating with engineering teams to redesign production-level execution workflows and evaluate infrastructure improvements for hedge funds, investment banks, proprietary trading firms, and other fintech applications

Software Engineering and Geoinformatics Intern Jun. 2025 – Aug. 2025

University of West Attica *Athens, Greece*

- Built 3 interactive virtual tours by processing 360° imagery into 3D maps and embedding location-aware content using HTML and geospatial software for use by 35 researchers.
- Designed early-stage real-time data evaluation software for collection and evaluation within virtual tours.

NASA CS & Research Intern Jun. 2023 – Sep. 2023

NASA, Center for Space Research *Austin, Texas*

- Developed Python scripts using NumPy and Pandas libraries to clean and process raw datasets from 10 universities, 3 telescopes and 500 stars, using AstroImageJ, and Stellarium to estimate open cluster age and distance.
- Presented research to ∼300 attendees at the NASA SEES Symposium; abstract accepted at AGU.

Software Engineer at the Liechtenstein Institute on Self-Determination Sep. 2025 – May 2026

Princeton University *Princeton, New Jersey*

- Developed and maintained the Institute's 25th Anniversary website using Drupal, HTML, and CSS, including an interactive historical timeline across 30 years. Collaborated on long-term digital initiatives.

Projects

Morgan Stanley Hackathon (2nd Place) | *Node.js, Flask, PostgreSQL, Supabase, Anthropic API* March 2026

- Built Grove, a platform aggregating 2,000+ NYC food resources to identify service gaps using geospatial and census data; Engineered full-stack, low-cost solution ($0 for demo, less than $20 for production) across 3 days, placing 2nd
- Developed a funding portfolio simulator and AI powered report generator using the Anthropic API to synthesize geospatial and census data into actionable insights for nonprofits and city planners.

TigerTA | *Python, Render, Neon, PostgreSQL, Jinja2, Psycopg* Feb. 2026 – May. 2026

- Led a 5-person team building a queue management platform for Princeton COS labs, supporting 100+ students and 30+ TAs.
- Engineered concurrent TA-student matching and real-time queue synchronization via RESTful APIs with normalized PostgreSQL schemas and role-based workflows, preventing race conditions.

Leadership

Tiger Game Developers (President) Mar. 2025 – Present

Princeton University *Princeton, New Jersey*

- Founded Princeton's game development club, organizing technical workshops, game jams, and collaborative software projects.